

PAUL KARONJI WAITHAKA

Full-Stack Software Engineer · Systems Architect · Founder

Nairobi, Kenya · paul-karonji.github.io/Portfolio · linkedin.com/in/paul-karonji-562080381 · github.com/paul-karonji

PROFESSIONAL SUMMARY

Full-stack software engineer and founder with 3+ years delivering production-grade systems across enterprise, SaaS, mobile, and AI domains. Final year student at KCA University (BSc Software Development). Founded WIK Technologies, shipping hardware-licensed POS systems, a property marketplace, AI agents, and consumer PWAs — all serving live users. Deep expertise in offline-first architecture, emerging-market payment integrations (M-Pesa, Paystack), and building complete product ecosystems from zero to production. Comfortable from Electron desktop to React Native mobile to multi-tenant Laravel SaaS.

EXPERIENCE

Founder & Lead Engineer | WIK Technologies

Oct 2025 – Present

Nairobi, Kenya

- ▶ Founded a custom software company delivering web, mobile, cloud, and AI solutions for clients across Kenya and globally.
- ▶ Architected and shipped enterprise POS suite (WIK-POS Premium & Standard) — Electron + React + TypeScript, SQLite offline DB, hardware-bound JWT licensing, WhatsApp receipts, customer loyalty CRM, and reorder logistics.
- ▶ Built and operate Pos-License-Server: centralized licensing authority with hardware fingerprinting (CPU + MAC), 3 tier model (lifetime / 30-day subscription / 5-day trial), and remote revocation.
- ▶ Developed DueSync (duesync.wiktechnologies.com) — Next.js 15 task-management SaaS with two-way Google Calendar sync, Pomodoro mode, timezone-aware push/email notifications, and Vercel Cron jobs.
- ▶ Shipped K-Track (ktrack.vercel.app) — commission & payment management SaaS with Paystack M-Pesa + card, USD→KES conversion, Cloudflare R2 file storage, real-time Socket.IO chat, and anti-tampering nonce validation.
- ▶ Managed full product lifecycle: requirements gathering, system architecture, development, CI/CD deployment, and client relationships.

Industrial Attachment — Technical Lead | JHUB Africa

Sep 2025 – Dec 2025

Nairobi, Kenya

- ▶ Led development team and architected the JHUB Africa Innovation Tracker (innovation.jhubafrica.com) — a full-stack project management platform for JKUAT Hub serving innovators, mentors, investors, and admins across Kenya.
- ▶ Built in PHP, MySQL, JavaScript, and Docker (Dockerized for production with docker-compose); 4 user roles (Admin, Mentor, Innovator, Investor), 6-stage project lifecycle tracking, mentor assignment engine, threaded comment system, document management, resource library, automated email notifications, and full activity audit log.
- ▶ Conducted code reviews, established team coding standards (PSR-12), and delivered the system on schedule — the platform is now live and in active use at JHUB Africa.

Freelance Software Developer | Independent

2023 – Present

Nairobi, Kenya (Remote)

- ▶ Delivered 10+ end-to-end web applications for diverse clients — from e-commerce and property management to agricultural advisory and hospital systems.
- ▶ Maintained 98% client satisfaction through clear communication, detailed documentation, and production-quality code with 90+ PageSpeed scores.
- ▶ Handled everything from requirements and architecture through deployment and ongoing maintenance with zero missed deadlines.

SELECTED PROJECTS

MtaaKeys — Property Marketplace & Rental Management Platform (yet to be publicly available)

Stack: Next.js 16, Express 5, TypeScript, PostgreSQL, Prisma, Socket.IO, Paystack, Africa's Talking, Cloudinary, Resend

- ▶ Full-stack monorepo (Next.js frontend / Express API / Vite admin portal) combining a public property marketplace with a complete rental management system — 5 user roles (Seeker, Agent, Landlord, Land Agent, Admin) with granular Row-Level Security.
- ▶ Dual Paystack subscription systems: tiered agent plans (FREE/STARTER/PROFESSIONAL/PREMIUM) gating building & unit quotas, plus a separate Land Agent tier (FREE/SILVER/GOLD/PLATINUM, up to KES 15,000/month). Pay-per-use listing Boosts and automated node-cron lifecycle jobs for renewals, expiry reminders, and tier downgrades.
- ▶ Algorithmic featured-listing curation (logarithmic view/inquiry scoring, recency decay, monopoly caps per agent/county/type); real-time Socket.IO messaging; Admin Audit Log with exact payload diffing (oldData vs newData); Africa's Talking SMS + Resend email notifications.
- ▶ Security hardening: JWT auth, CSRF, Helmet, rate limiting on auth routes, Paystack webhook signature verification against raw bytes, strict CORS whitelisting, and automated security invariant tests in GitHub Actions CI/CD preventing regressions.

Hiro — Self-Hosted AI Agent · Self-hosted (Telegram / WhatsApp)

Stack: Node.js, TypeScript, Multi-model LLM mesh, PostgreSQL

- ▶ Self-hosted AI agent with multi-model mesh workflows, parallel task execution, and a dynamic self-improving skills system.
- ▶ Supports document ingestion (PDF, DOCX), web search, memory-backed conversations, voice input/output, and browser canvas for rich output.

DueSync — Smart Task Management PWA · duesync.wikitechnologies.com

Stack: Next.js 15, TypeScript, PostgreSQL, Prisma, NextAuth v5, Upstash Redis, Vercel Cron, Resend, Google Calendar API v3, Radix UI, React Query, Zod

- ▶ Full-featured task management PWA and native Android app (TWA — Trusted Web Activity) with two-way Google Calendar sync (OAuth 2.0, proactive token refresh, event CRUD), recurring tasks, Pomodoro focus mode, timezone-aware smart reminders, daily digest emails (Resend), VAPID push notifications, and Vercel Cron jobs for background scheduling.
- ▶ Security hardened: RLS on all tables, strict CSP headers, CSRF protection, Upstash Redis rate limiting (auth: 5 req/15 min, calendar: 50 req/hr), Zod input validation, HTTP-only session cookies, and proactively patched CVE-2025-55182 / CVE-2025-66478 (Next.js RSC vulnerability).
- ▶ Ships a native MCP server (stdio + Streamable HTTP + Vercel route) exposing 13 tools for AI agent integration — enabling agents to list, create, update, complete tasks, and run a scoring-based day planner across available focus windows.

WIK-POS Premium — Enterprise Point of Sale · wikitechnologies.com

Stack: Electron, React, TypeScript, SQLite, JWT

- ▶ Hardware-bound licensing (CPU + MAC fingerprint), offline grace periods, customer loyalty CRM, WhatsApp e-receipts, and reorder logistics.
- ▶ Extensively documented (371 KB README with architecture diagrams); deployed to live retail clients.

Brixton Makunga HMS — Enterprise Hospital Management System · hmisdemo.wikitechnologies.com

Stack: Laravel 12, React 19, Vite, MySQL

- ▶ 200+ endpoint RESTful API, 56 database migrations, 35 Eloquent models, automated billing via Observer pattern, real-time queue management, and drug interaction checking.
- ▶ Multi-role staff access: Doctors, Nurses, Pharmacists, Lab Technicians, Receptionists. FEFO batch tracking in pharmacy module.

wikitrack — M-Pesa Income Tracker · React Native / Expo

Stack: React Native, Expo, Kotlin, Supabase, Firebase FCM

- ▶ Android app with native SMS reading via custom Kotlin module — parses M-Pesa SMS messages automatically.
- ▶ Local-first with Expo SQLite + background Supabase sync, Firebase FCM push notifications, and Vercel serverless backend.

K-Track — Multi-Tutor Agency & Commission Platform · ktrack.vercel.app

Stack: React 19, Vite, Node.js, Express 5, MySQL 8 / TiDB Serverless, Socket.IO, Cloudflare R2, Paystack, Resend

- ▶ Full-stack SaaS for commission-based tutor agencies: multi-role access (superadmin / tutor / client), task pooling and claiming, per-task real-time chat (Socket.IO), file delivery (Cloudflare R2), quoting, Paystack payments with USD→KES conversion, and referral/discount system with dynamically credited balances.
- ▶ Security-hardened auth: HMAC-SHA256 timing-safe guest checkout URLs (crypto.timingSafeEqual preventing side-channel attacks), HttpOnly refresh token cookies, rate limiting on all critical endpoints, and scoped file authorization restricting tutors strictly to their assigned tasks.
- ▶ Idempotent boot-time database patch service compatible with TiDB Serverless — auto-migrates schema, backfills legacy accounts, and adds missing tables on startup without manual migrations.

JHUB Africa Innovation Tracker · innovation.jhubafrica.com

Stack: PHP, MySQL, JavaScript, Docker, PHPMailer, SendGrid

- ▶ Full-stack project management platform for JKUAT Hub — 4 user roles (Admin, Mentor, Innovator, Investor), 6-stage project lifecycle (Activation → Mentorship → Capacity Building → Incubation → Showcase → Scale-Up), complete application workflow with approve/reject, and mentor assignment engine.
- ▶ Threaded comment system, document & pitch deck management, resource library, automated email notifications, comprehensive activity audit log, and investor portal for browsing live innovations — Dockerized for production with docker-compose.

Unsent — Private Diary PWA · unsent-eosin.vercel.app

Stack: React, Vite, VitePWA, Supabase, IndexedDB

- ▶ Fully offline-capable PWA: IndexedDB caching, background sync, Supabase RLS, local password hashing, auto-lock idle timer.
- ▶ Voice dictation (Web Speech API), media attachments, calendar view, 'On This Day' memory surfacing, full ZIP data export.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES6+), PHP, Python, Kotlin, HTML5, CSS3, SQL, Bash
Frontend	React, Next.js, React Native (Expo), Electron, Vite, Tailwind CSS, Radix UI, shadcn/ui, React Query (TanStack), React Hook Form, Framer Motion, PWA, Responsive Design
Backend	Node.js, Express, Laravel 10/12, Next.js API Routes, REST APIs, Socket.IO,
WebSockets	
Databases	PostgreSQL, MySQL, SQLite, Supabase, Firebase Firestore, Prisma ORM, Eloquent ORM
Cloud & DevOps	Vercel, Cloudflare R2, Firebase, Docker, GitHub Actions CI/CD, Vercel Cron, Upstash Redis, TiDB Serverless
Payments & APIs	M-Pesa Daraja (STK Push), Paystack, Google Calendar API v3, Africa's Talking, Cloudinary, OAuth2, Resend, VAPID Push Notifications, MCP (Model Context Protocol), PHPMailer
Security	JWT, Hardware fingerprinting, RLS, CSRF, Helmet, Rate limiting, Webhook verification, CSP headers, Zod validation, HMAC-SHA256 timing-safe signing, CVE patching
Architecture	Offline-first, multi-tenant SaaS, hardware licensing, monorepo, Observer pattern, event-driven
Testing & Docs	Jest (45+ tests), Swagger/OpenAPI, CI/CD security invariant testing, 400KB+ README documentation
Tools	Git/GitHub, Docker, Figma, Postman, VS Code

EDUCATION

BSc Software Development | KCA University

Sep 2021 – Present

Final Year Student · Nairobi, Kenya

Relevant Coursework: Advanced Web Development · Data Structures & Algorithms · Database Management · Network Security · Software Engineering · System Integration · Mobile Application Development · OOP · Agile Project Management

CERTIFICATIONS

- Cybersecurity Fundamentals — IBM SkillsBuild (2025)
- Data Protection Basics — Atingi eLearning Platform (Sep 2025)
- Zoho Creator Developer Certification — Zoho (Nov/Dec 2025)